ARQ, FEC, and HARQ Comparison Sheet

ARQ, FEC, and HARQ are three closely related error-control techniques used in digital and wireless communication systems. Their main purpose is to improve data transmission reliability over noisy channels.

Feature	FEC	ARQ	HARQ
Full Title	Forward Error Correction	Automatic Repeat reQuest	Hybrid Automatic Repeat reQuest
Basic Principle	Correct errors using redundant bits	Detect errors and request retransmission	Combine FEC and ARQ
Error Detection	Usually supported by coding or an additional CRC	Yes, typically CRC	Yes, typically CRC
Error Correction	Yes	No, not by ARQ itself	Yes
Retransmission	Not required	Required when errors are detected	Required when FEC decoding fails
Feedback (ACK/NACK)	Not necessary	Required	Required
Redundancy	Coding redundancy	Retransmitted packets	Coding redundancy and retransmissions
Delay	Generally predictable	Variable	Variable
Bandwidth Efficiency	Depends on code rate and channel	Decreases with retransmissions	Often better than conventional ARQ in noisy channels
Complexity	Encoder and decoder	Relatively low	Relatively high
Main Advantage	Corrects errors without waiting for feedback	Reliable delivery through retransmissions	Combines coding gain with retransmission reliability
Main Disadvantage	Redundancy even when the channel is good	Retransmission delay	Higher implementation complexity
Applications	Satellite, broadcasting, optical links	Data link protocols, reliable packet networks	LTE, 5G NR, modern wireless systems